

Complete Data Science & Analytics Mastery Syllabus

© 2026 Ajayeswaran Raja. All Rights Reserved.

Program Overview

This syllabus is designed to transform a beginner into an industry-ready:

- Data Analyst
- Business Analyst
- Data Scientist
- AI Engineer
- Machine Learning Engineer
- Analytics Consultant
- BI Developer
- Research Engineer

The syllabus covers:

- Data Analytics
 - Statistics
 - Python Programming
 - SQL
 - Data Visualization
 - Machine Learning
 - Deep Learning
 - Generative AI
 - MLOps
 - Cloud Analytics
 - Real-Time Projects
 - Interview Preparation
-

Phase 1 — Foundations of Data Science

Module 1: Introduction to Data Science

Topics

- What is Data Science?

- Data Science Lifecycle
- Types of Analytics
- Descriptive Analytics
- Diagnostic Analytics
- Predictive Analytics
- Prescriptive Analytics
- Roles in Data Science
- Data Science Applications
- AI vs ML vs DL vs Data Science
- Industry Use Cases

Hands-on

- Understanding datasets
 - Basic analytics workflow
-

Module 2: Mathematics for Data Science

Topics

- Basic Mathematics
- Linear Algebra
- Matrices
- Vectors
- Probability
- Statistics
- Calculus Basics
- Optimization Concepts

Important Concepts

- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Correlation
- Covariance
- Probability Distribution

Hands-on

- Statistical calculations using Python
-

Phase 2 — Python for Data Science

Module 3: Python Fundamentals

Topics

- Python Basics
- Variables
- Data Types
- Operators
- Conditional Statements
- Loops
- Functions
- OOP Concepts
- Exception Handling
- File Handling

Hands-on

- Student Data Analysis
 - Mini Analytics Programs
-

Module 4: Advanced Python for Analytics

Topics

- Lambda Functions
- List Comprehensions
- Generators
- Decorators
- Iterators
- Virtual Environments
- Package Management

Libraries

- math
 - os
 - sys
 - datetime
 - collections
-

Phase 3 — Data Handling & Processing

Module 5: NumPy

Topics

- Arrays
- Array Operations
- Indexing
- Slicing
- Broadcasting
- Matrix Operations
- Random Module
- Linear Algebra

Hands-on

- Matrix Calculations
 - Numerical Analysis
-

Module 6: Pandas

Topics

- Series
- DataFrames
- Reading CSV/Excel/JSON
- Data Cleaning
- Handling Missing Values
- Data Transformation
- Filtering
- Grouping
- Merging
- Aggregation
- Feature Engineering

Hands-on

- Employee Data Analysis
 - Sales Data Processing
-

Module 7: Data Wrangling

Topics

- Data Cleaning
- Data Transformation
- Encoding Techniques
- Handling Outliers
- Scaling
- Normalization
- Feature Selection
- Data Pipelines

Hands-on

- End-to-End Data Cleaning Project
-

Phase 4 — SQL for Data Analytics

Module 8: SQL Fundamentals

Topics

- Database Basics
- SQL Queries
- CRUD Operations
- Constraints
- Joins
- Subqueries
- Views
- Stored Procedures
- Indexing

Databases

- MySQL
- PostgreSQL
- SQLite

Hands-on

- Employee Database Analysis
 - E-commerce Database Queries
-

Module 9: Advanced SQL for Analytics

Topics

- Window Functions
- Common Table Expressions
- Analytical Queries
- Query Optimization
- Data Warehousing Basics
- ETL Concepts

Hands-on

- Dashboard Backend Queries
-

Phase 5 — Data Visualization

Module 10: Data Visualization Fundamentals

Topics

- Visualization Principles
 - Storytelling with Data
 - Charts and Graphs
 - KPI Design
 - Dashboard Design
-

Module 11: Matplotlib & Seaborn

Topics

- Line Charts
- Bar Charts
- Pie Charts
- Histograms
- Scatter Plots
- Heatmaps
- Pairplots
- Distribution Plots

Hands-on

- Analytics Dashboard
-

Module 12: Power BI & Tableau

Topics

- Power BI Basics
- Data Modeling
- DAX
- Dashboard Building
- Tableau Visualization
- Interactive Reports
- Business Dashboards

Hands-on

- HR Analytics Dashboard
 - Sales Analytics Dashboard
-

Phase 6 — Statistics & Probability

Module 13: Statistics for Data Science

Topics

- Descriptive Statistics
- Inferential Statistics
- Hypothesis Testing
- Confidence Intervals
- Sampling
- Correlation Analysis
- ANOVA
- Chi-Square Test

Hands-on

- Statistical Analysis Projects
-

Module 14: Probability for Machine Learning

Topics

- Probability Rules
- Bayes Theorem
- Conditional Probability
- Probability Distributions

- Normal Distribution
- Bernoulli Distribution
- Binomial Distribution
- Poisson Distribution

Hands-on

- Probability Simulations
-

Phase 7 — Machine Learning

Module 15: Introduction to Machine Learning

Topics

- ML Workflow
 - Types of ML
 - Supervised Learning
 - Unsupervised Learning
 - Reinforcement Learning
 - Train/Test Split
 - Model Evaluation
 - Bias-Variance Tradeoff
-

Module 16: Supervised Learning

Algorithms

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- SVM
- KNN
- Naive Bayes
- Gradient Boosting
- XGBoost

Metrics

- Accuracy
- Precision
- Recall
- F1 Score

- ROC-AUC

Projects

- House Price Prediction
 - Employee Attrition Prediction
-

Module 17: Unsupervised Learning

Algorithms

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN
- Gaussian Mixture Models
- PCA
- t-SNE
- UMAP

Projects

- Customer Segmentation
 - Market Basket Analysis
-

Phase 8 — Deep Learning

Module 18: Neural Networks

Topics

- Perceptrons
- Activation Functions
- Forward Propagation
- Backpropagation
- Loss Functions
- Optimizers

Frameworks

- TensorFlow
 - Keras
 - PyTorch Basics
-

Module 19: Computer Vision

Topics

- CNN Architecture
- Image Processing
- Object Detection
- Transfer Learning
- OpenCV

Projects

- Face Detection
 - Image Classification
 - Handwritten Digit Recognition
-

Module 20: Natural Language Processing

Topics

- Text Preprocessing
- Tokenization
- Stemming
- Lemmatization
- TF-IDF
- Word Embeddings
- Transformers
- BERT
- GPT Concepts

Projects

- Sentiment Analysis
 - Chatbot Development
-

Phase 9 — Generative AI

Module 21: Generative AI Fundamentals

Topics

- Introduction to GenAI
- LLM Architecture
- Prompt Engineering

- RAG Systems
- Vector Databases
- Embeddings
- AI Agents
- LangChain
- Fine Tuning

Models

- GPT
- Gemini
- Claude
- Llama

Projects

- AI Resume Generator
- AI Tutor
- AI Document Analyzer

Phase 10 — Big Data & Cloud Analytics

Module 22: Big Data Fundamentals

Topics

- Hadoop Ecosystem
- Spark Basics
- Distributed Computing
- Data Lakes
- Streaming Analytics

Hands-on

- PySpark Processing

Module 23: Cloud for Data Science

Platforms

- AWS
- Azure
- Google Cloud

Topics

- S3
- EC2
- SageMaker
- BigQuery
- Azure ML
- Data Pipelines

Projects

- Cloud Analytics Pipeline
-

Phase 11 — MLOps & Deployment

Module 24: MLOps

Topics

- Model Deployment
- Docker
- Kubernetes Basics
- CI/CD for ML
- MLflow
- DVC
- Monitoring ML Models

Hands-on

- Deploy ML Models using FastAPI
-

Module 25: API Development for ML

Topics

- Flask APIs
- FastAPI
- REST APIs
- Model Serving
- Swagger Documentation

Projects

- ML Prediction API

Phase 12 — Business Analytics

Module 26: Business Analytics

Topics

- KPI Analysis
- Business Metrics
- Forecasting
- A/B Testing
- Customer Analytics
- Financial Analytics
- Marketing Analytics

Projects

- Retail Analytics Dashboard
 - Business Intelligence Reports
-

Phase 13 — Time Series Analysis

Module 27: Time Series Forecasting

Topics

- Time Series Basics
- Trend Analysis
- Seasonality
- ARIMA
- SARIMA
- Facebook Prophet
- LSTM for Forecasting

Projects

- Stock Price Prediction
 - Sales Forecasting
-

Phase 14 — Real-Time Projects

Beginner Projects

- Student Data Analysis
- Weather Analytics
- Expense Tracker
- CSV Analyzer

Intermediate Projects

- Sales Dashboard
- HR Analytics
- Recommendation System
- Customer Segmentation

Advanced Projects

- AI Chatbot
 - Real-time Analytics Platform
 - Fraud Detection System
 - Traffic Prediction System
 - Intelligent Business Dashboard
 - GenAI Knowledge Assistant
-

Phase 15 — Interview Preparation

Module 28: Interview Preparation

Topics

- SQL Interview Questions
- Python Interview Questions
- Statistics Interview Questions
- ML Interview Questions
- Case Studies
- Business Problems
- Coding Challenges

Platforms

- LeetCode
- HackerRank
- Kaggle

Phase 16 — Research & Advanced Topics

Module 29: Advanced AI Topics

Topics

- Reinforcement Learning
- GANs
- Diffusion Models
- AI Ethics
- Explainable AI
- Federated Learning
- Multi-Agent Systems

Recommended Tools

Programming

- Python
- Jupyter Notebook
- VS Code

Databases

- MySQL
- PostgreSQL
- MongoDB

Visualization

- Power BI
- Tableau
- Plotly

Cloud

- AWS
- Azure
- Google Cloud

MLOps

- Docker
 - Kubernetes
 - MLflow
-

Recommended Learning Path

Beginner Level (1–2 Months)

- Python
- SQL
- Statistics
- Data Visualization

Intermediate Level (3–5 Months)

- Machine Learning
- Data Analytics
- Dashboard Development

Advanced Level (6–12 Months)

- Deep Learning
 - Generative AI
 - MLOps
 - Cloud Analytics
 - Research Projects
-

Certifications

Recommended Certifications

- Google Data Analytics
 - IBM Data Science Professional Certificate
 - Microsoft Power BI Certification
 - AWS Machine Learning Specialty
 - TensorFlow Developer Certificate
 - Databricks Certification
-

Outcome After Completion

After completing this syllabus, learners can work as:

- Data Analyst
 - Data Scientist
 - ML Engineer
 - AI Engineer
 - Business Analyst
 - BI Developer
 - Research Scientist
 - Analytics Consultant
 - AI Product Engineer
-

Final Advice

To become successful in Data Science:

1. Master Python and SQL deeply
 2. Learn statistics properly
 3. Build real-world projects
 4. Focus on problem solving
 5. Practice data storytelling
 6. Learn cloud and deployment
 7. Build GitHub portfolio projects
 8. Participate in Kaggle competitions
 9. Learn AI and Generative AI tools
 10. Stay updated with latest AI trends
-

Best Practice Strategy

Daily Routine

- 2 Hours Theory
- 3 Hours Hands-on Practice
- 1 Hour Problem Solving
- 1 Hour Project Building

Weekly Routine

- Build one mini project
- Solve SQL problems

- Practice ML algorithms
 - Upload work to GitHub
 - Write technical blogs
-

Final Capstone Projects

Enterprise-Level Projects

- AI-powered Analytics Dashboard
- Intelligent Fraud Detection System
- Recommendation Engine
- Predictive Maintenance System
- GenAI Business Assistant
- Real-time Streaming Analytics Platform
- AI Recruitment Analytics System
- Smart Healthcare Prediction System
- Financial Forecasting Platform
- Customer Churn Prediction System